CAO-DF - Activity Diagram: end-to-end study workflow (script per step)

Setup

Record machine configuration (environment_report.py)

Check imports, Java, Spark session, pipeline (verify_phase2_setup.py)

Data

Generate study workloads, seed 42 (workload_generator.py)

E1: shuffle-volume gate (verify_workloads.py)

Experiments

E2: exploratory partition sweep (explore_partition_curve.py)

E3: formal A1 matrix, 8 settings x 4 workloads x 5 blocks = 160 runs (run_a1_experiments.py)

E4: A2 pilot gate, 3 blocks (run_a2_experiments.py --pilot)

E5: formal A2 matrix, 6 workloads x 3 stages x 5 blocks = 90 runs (run_a2_experiments.py)

Analysis

Trimmed outlier check on A1 (robustness_check.py)

Cost model: P1 / P2 costs, N*, TC at N = 1, 5, 10, 50, price x0.5 / x1 / x1.5 (cost_model.py)

Data

Generate 4 unseen validation workloads (workload_generator.py --validation)

Experiments

E6: validation, 4 workloads x 10 stages x 3 blocks = 120 runs (run_validation.py)

Analysis

Decision rule + validation against static, performance-only and oracle; leave-one-out (decision_framework.py)

Paper figures (make_figures.py)

Paper

Write results from processed CSVs only

Compile IEEEtran paper (8 pages)